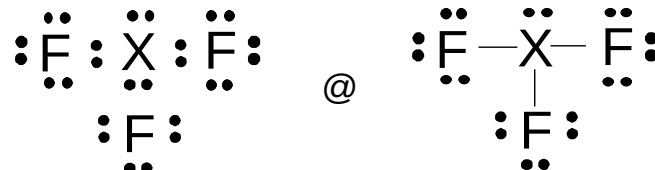
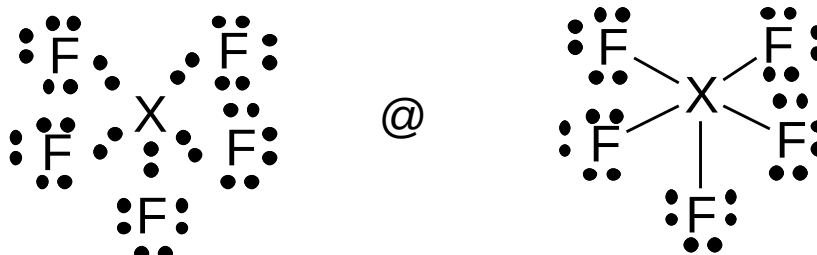
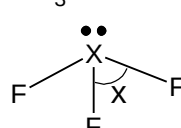
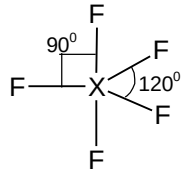


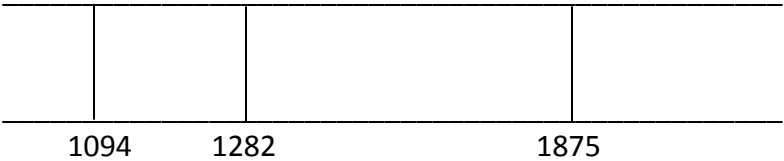
NO	PART	SCHEME																
1	(a)	Mass of C = $0.558 \times \frac{12}{44} = 0.15218g$																
		Mass of H = $0.344 \times \frac{2}{18} = 0.03822g$																
		Mass of N = $0.250 - (0.15218+0.03822) = 0.0596g$																
		<table><tr><th>Element</th><th>Carbon</th><th>Hydrogen</th><th>Nitrogen</th></tr><tr><td>Mass,g</td><td>1.15218</td><td>0.03822</td><td>0.0596</td></tr><tr><td>Number of Moles, mol</td><td>$\frac{0.15218}{12}$ = 0.01268</td><td>$\frac{0.03822}{1}$ = 0.03822</td><td>$\frac{0.0596}{14}$ = 0.00426</td></tr><tr><td>Smallest Ratio</td><td>$\frac{0.01268}{0.00426}$ = 2.97 ≈ 3</td><td>$\frac{0.03822}{0.00426}$ = 8.97 ≈ 9</td><td>$\frac{0.00426}{0.00426}$ = 1</td></tr></table>	Element	Carbon	Hydrogen	Nitrogen	Mass,g	1.15218	0.03822	0.0596	Number of Moles, mol	$\frac{0.15218}{12}$ = 0.01268	$\frac{0.03822}{1}$ = 0.03822	$\frac{0.0596}{14}$ = 0.00426	Smallest Ratio	$\frac{0.01268}{0.00426}$ = 2.97 ≈ 3	$\frac{0.03822}{0.00426}$ = 8.97 ≈ 9	$\frac{0.00426}{0.00426}$ = 1
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Empirical Formula of Q is C₃H₉N																		
	(b)	Molar mass of Q = $3(12) + 9(1) + 1(14)$ = 59.0 g mol⁻¹ (unit insist)																
	(c)	1 mol C ₃ H ₉ N $\equiv$ 9 mol H atom Mol of Q = $\frac{0.25}{59} = 4.24 \times 10^{-3}$ mol No of H atoms = $9 \times 4.24 \times 10^{-3} \times 6.02 \times 10^{23}$ = 2.295×10^{22} atoms @ 2.30×10^{22} atoms																

NO	PART	SCHEME		
2	(a)(i)	XF_3 $\text{X} = 5\text{e}$ $3\text{xF} = 3 \times 7\text{e} = 21\text{e}$ $\text{Total} = 5\text{e} + 21\text{e} = 26\text{e}$		
				
	(a)(ii)	XF_5 $\text{X} = 5\text{e}$ $5\text{xF} = 5 \times 7\text{e} = 35\text{e}$ $\text{Total} = 5\text{e} + 35\text{e} = 40\text{e}$		
				
		Molecule	Electron pair geometry @ Basic geometry	Molecular Geometry
		XF_3	Tetrahedral	Trigonal pyramidal
		XF_5	Trigonal Bipyramidal	Trigonal bipyramidal
	(a)(iii)	XF_3  $103^\circ \leq x < 109.5^\circ @ 109^\circ$ XF_5  Correct geometry Each bond angle are labelled		
	(b)	$\text{XF}_3 = \text{sp}^3$ $\text{XF}_5 = \text{sp}^3\text{d}$		

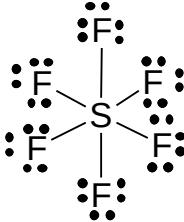
NO	PART	SCHEME
3	(a)(i)	Total mass of sample = 15.50g Mass of methane gas in the sample = $(82/100) \times 15.50 = 12.71\text{g}$ Mol of methane gas in the sample = $12.71 / 16 = 0.79 \text{ mol}$ Mass of ethane gas in the sample = $(18/100) \times 15.50 = 2.79\text{g}$ Mol of ethane gas in the sample = $2.79 / 30 = 0.09 \text{ mol}$ Total mol of gases in the sample = $0.79 + 0.09 = 0.88 \text{ mol}$
	(a)(ii)	$PV = nRT \text{ @ } P = \frac{nRT}{V}$ $= \frac{(0.88 \text{ mol})(0.08206 \text{ L atm mol}^{-1} \text{K}^{-1})(303.15 \text{ K})}{15 \text{ L}}$ P_{total} = 1.41 atm (unit insist)
	(a)(iii)	$P_{\text{methane}} = X_{\text{methane}} \times P_{\text{total}}$ $= \frac{0.79}{0.88} \times 1.41$ = 1.27 atm (unit insist) P_{ethane} = $1.41 - 1.27 = 0.14 \text{ atm}$ (unit insist)
	(b)(i)	C₃H₈ has larger size @ mass @ molar mass @ surface area than CH₄ Therefore C₃H₈ experience stronger Van der Waals forces @ London forces @ dispersion forces. H₂O can form hydrogen bond between their molecules while H₂S experience dipole-dipole forces. Hydrogen bond is stronger than dipole-dipole forces
	(b)(ii)	Solid CO₂ is a molecular covalent solids @ covalent molecule @ discrete @ simple covalent molecule in which each molecules is held by weak Van der Waals forces. Diamond is a giant/infinite/network covalent solids in which the solid is bound by a strong covalent bonds.

NO	PART	SCHEME																				
4	(a)(i)	Standard solution is a solution of accurately known concentration Equivalent point is the point where the mole of acid and base is stoichiometrically equivalent @ mole of H ⁺ = mole of OH ⁻ End point is the point where an indicator changes colour.																				
	(a)(ii)	pH = 7																				
	(a)(iii)	M _{NaOH} V _{NaOH} = M _{HCl} V _{HCl} M _{NaOH} = $\frac{22.50\text{ml} \times 1.0\text{M}}{20.00\text{ml}}$ = 1.12 M @ 1.125 M (unit insist)																				
	(a)(iv)	<u>After 25ml of HCl added</u> Mol acid added = 1.0 mol L ⁻¹ x 0.025 L = 0.025 mol Mol of NaOH = 1.12 mol L ⁻¹ x 0.020 L = 0.022 mol @ 1.125 mol L ⁻¹ x 0.020 L = 0.0225 mol Mol of excess HCl = 0.025 – 0.022 = 0.0030 mol @ 0.025 – 0.0225 = 0.0025 mol [HCl] = $\frac{0.0030 \text{ mol}}{0.045 \text{ L}}$ = 0.067 M @ $\frac{0.0025 \text{ mol}}{0.045 \text{ L}}$ = 0.0555 M pH = - log [0.067] = 1.17 @ 1.2 @ pH = -log[0.0555] = 1.26																				
	(b)	$\text{C}_5\text{H}_5\text{N (aq)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{C}_5\text{H}_5\text{NH}^+ \text{(aq)} + \text{OH}^- \text{(aq)}$ <table><tr><td></td><td>C₅H₅N (aq)</td><td>H₂O(l)</td><td>C₅H₅NH⁺ (aq)</td><td>OH⁻(aq)</td></tr><tr><td>Initial (M)</td><td>1.0</td><td>-</td><td>0</td><td>0</td></tr><tr><td>Change (M)</td><td>- x</td><td>-</td><td>+x</td><td>+x</td></tr><tr><td>Equilibrium (M)</td><td>1.0-x</td><td>-</td><td>x</td><td>x</td></tr></table> $K_b = \frac{[\text{C}_5\text{H}_5\text{NH}^+][\text{OH}^-]}{[\text{C}_5\text{H}_5\text{N}]} = \frac{[x][x]}{[1.0-x]} = 1.50 \times 10^{-9}$ Since K _b is very small, so x <<<<<<< 1.0 M @ 1.0 – x ≈ 1.0		C ₅ H ₅ N (aq)	H ₂ O(l)	C ₅ H ₅ NH ⁺ (aq)	OH ⁻ (aq)	Initial (M)	1.0	-	0	0	Change (M)	- x	-	+x	+x	Equilibrium (M)	1.0-x	-	x	x
	C ₅ H ₅ N (aq)	H ₂ O(l)	C ₅ H ₅ NH ⁺ (aq)	OH ⁻ (aq)																		
Initial (M)	1.0	-	0	0																		
Change (M)	- x	-	+x	+x																		
Equilibrium (M)	1.0-x	-	x	x																		

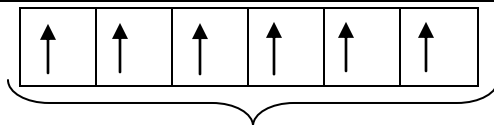
		$K_b = \frac{x^2}{0.20}$ $x = \sqrt{1.50 \times 10^{-9}} = 3.87 \times 10^{-5} \text{ @}$ $[\text{OH}^-] = 3.87 \times 10^{-5} \text{ M}$ $\text{pOH} = -\log [3.87 \times 10^{-5}] = \mathbf{4.41}$ $\text{pH} = 14 - 4.41 = 9.59$ pH of the solution is 9.59
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NO	PART	SCHEME
5	(a)	For Paschen Series, $n_1 = 3 \text{ @ } n_f = 3$ The first three possible transitions: $n=6 \rightarrow n=3$; $n=5 \rightarrow n=3$; $n=4 \rightarrow n=3$ $\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), n_1 < n_2$ $n=6 \rightarrow n=3; \frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{9} - \frac{1}{36} \right)$ $\lambda = 1094 \text{ nm @ } 1.094 \times 10^{-6} \text{ m}$ $n=5 \rightarrow n=3; \frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{9} - \frac{1}{25} \right)$ $\lambda = 1282 \text{ nm @ } 1.282 \times 10^{-6} \text{ m}$ $n=4 \rightarrow n=3; \frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{9} - \frac{1}{16} \right)$ $\lambda = 1875 \text{ nm @ } 1.875 \times 10^{-6} \text{ m}$ Line spectrum for this series:  An electron at the excited state @ ($n=6$, $n=5$ or $n=4$) is unstable and will fall back to $n=3$ @ lowest energy. Transitions from a higher energy level to a lower energy level release specific amount of energy @ specific wavelength @ specific frequency

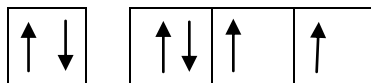
	(b)	Going down the group, the atom/ions increase in size Going down the group, the principle quantum number @ electron shell increases causing distance of valence electrons from the nucleus to increase @ shielding effect getting larger Nuclear charge are the same for the species with its corresponding ions, but the negative ions have a larger radius than the neutral atom because more repulsion among electrons (one more than the neutral atom) and are held less tightly to the nucleus @ force of attraction between nucleus and the outermost electrons are getting weaker Ionization energy decreases going down the group The size of atom is larger going down the group. Therefore, outermost electron is located further away from nucleus @ weaker attraction between nucleus and outermost electron. Therefore, less energy is needed @ easy to remove electron. Electron affinity decreases going down the group. The size of the atom is larger going down the group. Addition of electron causes weaker attraction of electrons towards nucleus @ easier to lose electron.

NO	PART	SCHEME																										
6		Valence Bond Theory Lewis structure of SF₆:  <u>Hybridisation of sulphur in SF₆</u> S (ground state): <table style="margin-left: auto; margin-right: auto;"><tr><td style="border: 1px solid black; padding: 5px; text-align: center;">↑↓</td><td colspan="3" style="border: 1px solid black; padding: 5px; text-align: center;">↑↓ ↑ ↑</td></tr><tr><td style="text-align: center;">3s</td><td colspan="3" style="text-align: center;">3p</td></tr></table> S (excited state): <table style="margin-left: auto; margin-right: auto;"><tr><td style="border: 1px solid black; padding: 5px; text-align: center;">↑</td><td style="border: 1px solid black; padding: 5px; text-align: center;">↑</td><td style="border: 1px solid black; padding: 5px; text-align: center;">↑</td><td style="border: 1px solid black; padding: 5px; text-align: center;">↑</td><td style="border: 1px solid black; padding: 5px; text-align: center;">↑</td><td style="border: 1px solid black; padding: 5px; text-align: center;">↑</td><td style="border: 1px solid black; padding: 5px; text-align: center;"> </td><td style="border: 1px solid black; padding: 5px; text-align: center;"> </td><td style="border: 1px solid black; padding: 5px; text-align: center;"> </td></tr><tr><td style="text-align: center;">3s</td><td colspan="3" style="text-align: center;">3p</td><td colspan="5" style="text-align: center;">3d</td></tr></table>	↑↓	↑↓ ↑ ↑			3s	3p			↑	↑	↑	↑	↑	↑				3s	3p			3d				
↑↓	↑↓ ↑ ↑																											
3s	3p																											
↑	↑	↑	↑	↑	↑																							
3s	3p			3d																								

S (hybrid state):

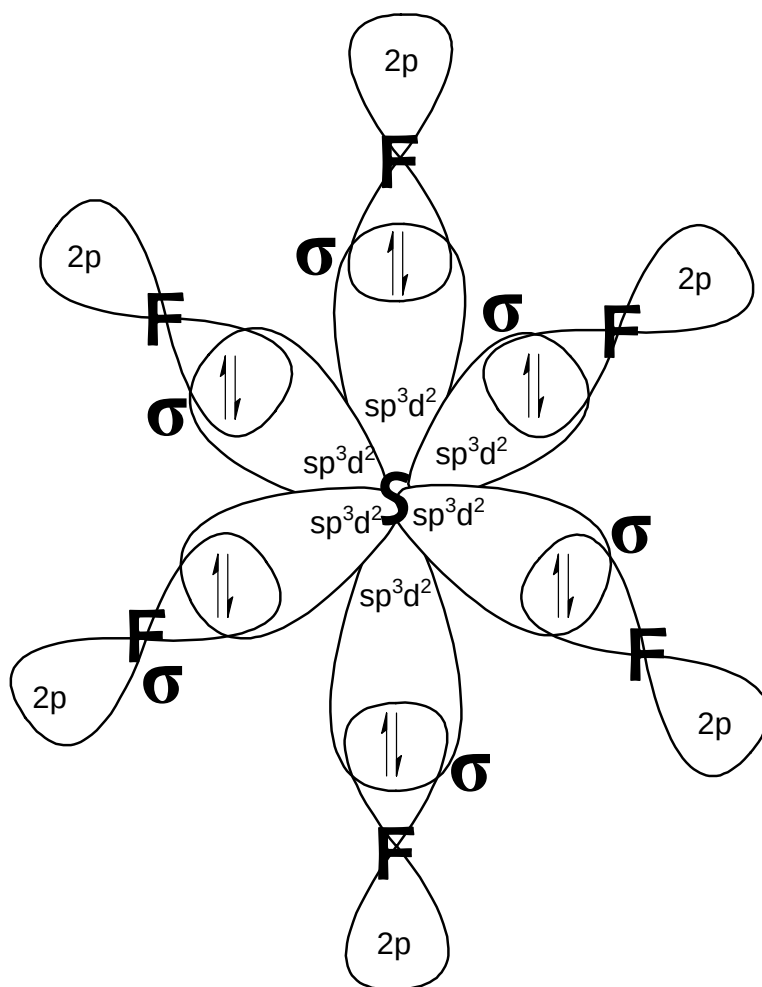
 sp^3d^2

F (ground state):



2s

2p



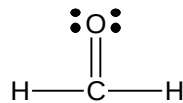
Correct geometry octahedral

Correct overlapping to form sigma bond

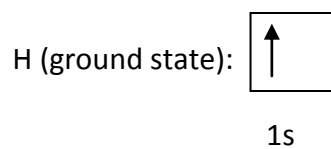
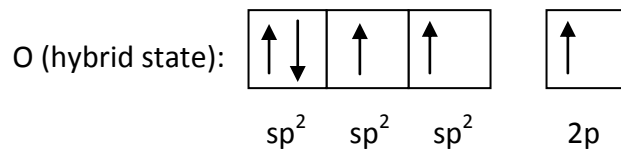
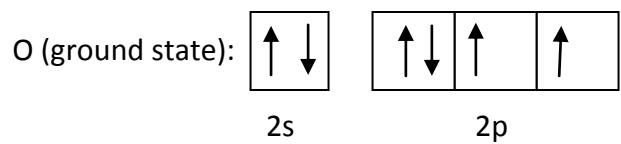
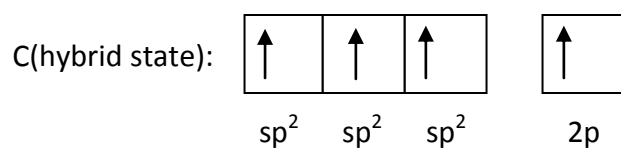
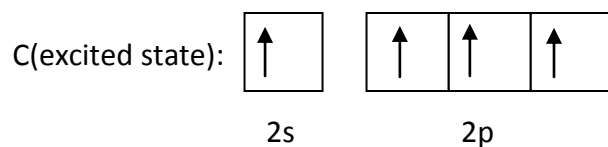
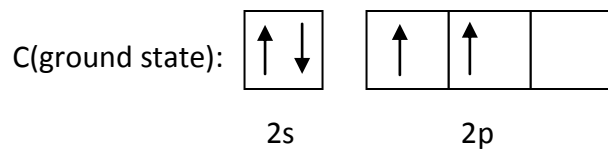
Complete label

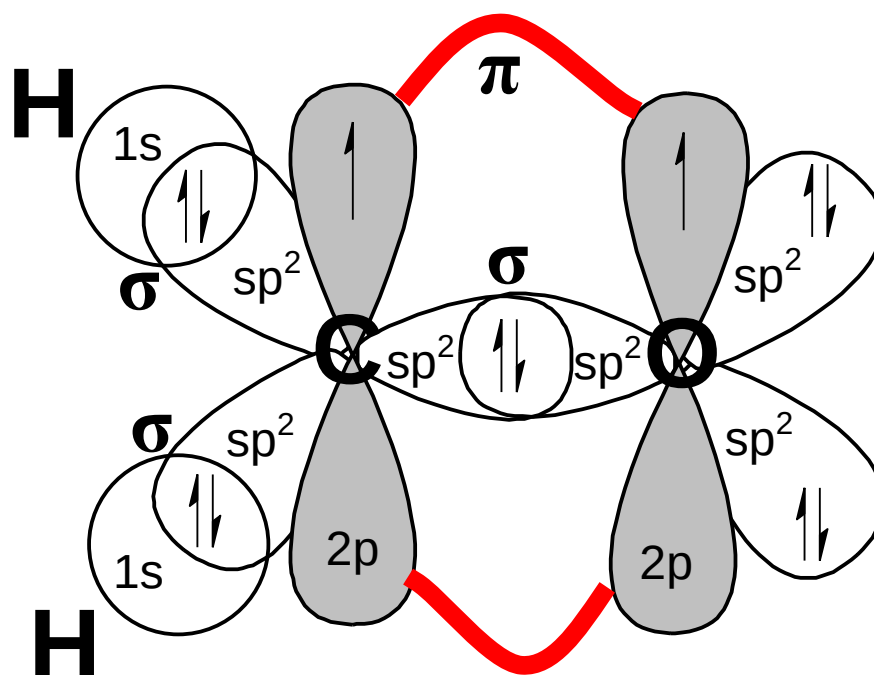
Bond angle 90°

Lewis Structure of H_2CO :



Hybridisation of sulphur in H_2CO



Orbital overlaps for H₂CO

Correct geometry, trigonal planar

Correct overlapping to form sigma bond (3 sigma bonds)

Correct overlapping to form pi bond (1 pi bond)

Complete label

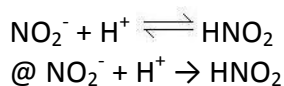
Bond angle 120°

NO	PART	SCHEME																
7	(a)	Ionic Cation and anion held by electrostatic interaction or ionic bond Example: NaCl Molecular Covalent Molecule forms lattice by weak Van der Waals Example: CO₂ (dry ice), H₂O(s), S₈, P₄, I₂, O₂ Metallic Attraction between positively charged metal ion surrounded by sea of electrons Example: Na/Mg/Cu Giant covalent Nonmetal atom held together by electron sharing/covalent bond to form giant structure Example: Diamond/graphite/SiO₂/Si																
	(b)	At 600 K $2\text{NO} + \text{O}_2 \rightleftharpoons 2\text{NO}_2, K_c = 7.7 \times 10^7$ $[\text{NO}] = [\text{O}_2] = [\text{NO}_2] = \frac{2.0 \times 10^{-3} \text{ mol}}{2\text{L}} = 1.0 \times 10^{-3} \text{ M}$ $Q = \frac{[\text{NO}_2]^2}{[\text{NO}]^2 [\text{O}_2]} @ \frac{[1.0 \times 10^{-3}]^2}{[1.0 \times 10^{-3}]^2 [1.0 \times 10^{-3}]} @ 1.0 \times 10^3$ Since Q ≠ K_c @ Q < K_c the reaction is not in equilibrium																
		Since [O₂] at 450K (9.6 × 10⁻⁴ M) less than that at 600K (1.0 × 10⁻³ M), the equilibrium position moves to the right $2\text{NO} + \text{O}_2 \rightleftharpoons 2\text{NO}_2$ <table><tr><td></td><td>NO</td><td>O₂</td><td>NO₂</td></tr><tr><td>Initial []</td><td>0.001</td><td>0.001</td><td>0.001</td></tr><tr><td>Δ</td><td>-2x</td><td>-x</td><td>+2x</td></tr><tr><td>[] eq</td><td>0.001 – 2x</td><td>0.001 - x</td><td>0.001 + 2x</td></tr></table> If the [O₂] at equilibrium is 9.6 × 10⁻⁴ M Thus, 0.001 – x = 9.6 × 10⁻⁴ x = 0.001 – 9.6 × 10⁻⁴ @ 4.0 × 10⁻⁵ M [NO₂] = 0.001 + (2 × 4.0 × 10⁻⁵) = 0.001 + 8.0 × 10⁻⁵ @ 1.08 × 10⁻³ M		NO	O₂	NO₂	Initial []	0.001	0.001	0.001	Δ	-2x	-x	+2x	[] eq	0.001 – 2x	0.001 - x	0.001 + 2x
	NO	O₂	NO₂															
Initial []	0.001	0.001	0.001															
Δ	-2x	-x	+2x															
[] eq	0.001 – 2x	0.001 - x	0.001 + 2x															

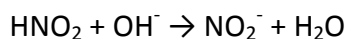
		$[\text{NO}] = 0.001 - (2 \times 4.0 \times 10^{-5})$ $= 0.001 - 8.0 \times 10^{-5} \text{ @ } 9.2 \times 10^{-4} \text{ M}$ $\text{At 450K, } K_c = \frac{(1.08 \times 10^{-3})^2}{(9.2 \times 10^{-4})^2 (9.6 \times 10^{-4})}$ $= 1.43 \times 10^3$
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NO	PART	SCHEME
8	(a)	$\text{Mg(OH)}_2(\text{s}) \rightleftharpoons \underset{x}{\text{Mg}^{2+}(\text{aq})} + \underset{2x}{2\text{OH}^{-}(\text{aq})} \text{ (phase insist)}$ $K_{sp} = [\text{Mg}^{2+}][\text{OH}^{-}]^2 \text{ @ } (x)(2x)^2$ $4x^3 = 8.9 \times 10^{-12}$ $x = 1.31 \times 10^{-4}$ $[\text{Mg}^{2+}] = 1.31 \times 10^{-4} \text{ M (unit insist)}$
		In saturated Mg(OH)_2 solution containing 0.01M NaOH $[\text{OH}^{-}] = (0.01 + 2x) \text{ M}$ X is too small thus $[\text{OH}^{-}]$ is assumed to be 0.01 M $K_{sp} = [\text{Mg}^{2+}] \times [0.01]^2$ $[\text{Mg}^{2+}] = (8.9 \times 10^{-12}) / (0.01)^2$ $[\text{Mg}^{2+}] = 8.9 \times 10^{-8} \text{ M (unit insist)}$
	(b)	$[\text{NO}_2^{-}] = 0.125/0.5 \text{ @ } 0.25\text{M}$ $[\text{HNO}_2] = 0.25\text{M}$ $\text{HNO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{NO}_2^{-} + \text{H}_3\text{O}^{+}$ $\text{pH} = \text{pK}_a - \log ([\text{HNO}_2]/[\text{NO}_2^{-}])$ $= 3.29 - \log (0.25/0.25)$ $= 3.29$

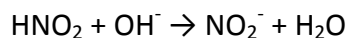
When a small amount of strong acid is added, NO_2^- will react with the added H^+ to form **weak acid HNO_2** , since the **dissociation is negligible** (K_a is very small), the pH is maintained.



When a small amount of strong base is added, **HNO_2** will react with the **added OH^- to form H_2O** thus the pH is maintained



Addition of 0.001 mol NaOH



	HNO_2	OH^-	NO_2^-	H_2O
n_i	0.125	0.001	0.125	-
Δ	-0.001	-0.001	+0.001	-
n_f	0.125 - 0.001 @ 0.124	0.001 - 0.001	0.125 + 0.001 @ 0.126	-
[]	0.124 / 0.5 @ 0.248		0.126 / 0.5 @ 0.252	-

$$\text{pH} = \text{p}K_a - \log ([\text{HNO}_2]/[\text{NO}_2^-])$$

$$= 3.29 - \log (0.248/0.252)$$

$$= 3.29 + 6.9 \times 10^{-3}$$

$$= 3.30$$

$$\text{Change in pH value} = 3.30 - 3.29$$

$$= \mathbf{0.01}$$